

Part A — Design Process (what your teacher expects you to “define + explain”)

1) Overview of design stages (concept → physical → tools/standards)

Design of an interactive system is **multi-stage** and **iterative**: you start from a high-level concept, translate it into concrete UI, and keep evaluating throughout.

a) Technology Myopia (VERY testable)

Technology myopia = focusing too narrowly on tech features while ignoring the **human + social + messy real context** of use. Your lectures emphasize designers must remain sensitive to:

- **Human-human communication** (tech often mediates people-to-people interaction; design must support it)
- **Implicit/tacit knowledge** (real work includes unspoken practices users can't easily explain)
- **Non-technical realities** (emotions, politics, workarounds; ignoring these makes systems fail)

Exam line: “A purely technical solution that ignores the real work context is likely to fail.”

b) Conceptual Design (what it includes)

This is the “**what is the system**” level:

- **Structuring the information space** (organize data/functions logically)
- **Creating alternative solutions** (brainstorm multiple concepts)
- **Choosing the concept to pursue** (evaluate alternatives vs user needs + project goals)

c) Physical Design (what it includes)

This converts concept into an actual UI:

- Visual look: **branding, color palette, typography**
- Components: **buttons, menus, forms, nav bars**
- Screen-by-screen **layout + workflow**

d) Standards & tools (why they matter)

Physical design is supported by:

- **Graphical libraries/UI toolkits** (e.g., Material Design, Apple HIG) for consistency

- Visual builders / web dev tools that help construct UI and may generate code
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Part B — Evaluation & Measurement (traditional vs modern, methods, triangulation)

2) Measuring the user experience

Traditional usability measurements (“classic trifecta”)

1. **Task performance / effectiveness:** can users complete tasks? (e.g., % tasks completed)
2. **Time performance / efficiency:** how long does it take?
3. **User satisfaction:** how users feel (questionnaires like **SUS**)

Why traditional metrics are NOT enough anymore

Lectures explicitly say these miss modern UX realities like:

- **Why users abandon** an interface
- **Discretionary use** (using for enjoyment: YouTube/games)
- Human factors like **emotion + trust**

Newer measurements (what to mention in answers)

Modern evaluation adds:

- Feelings like **community, enjoyment**, emotional engagement
- Satisfaction from **accomplishments in virtual worlds**
- **Physiological measures** like **EEG** (cognitive load) and **EMG** (muscle tension)

Your doc also literally flags: “Traditional metrics miss emotion + trust + enjoyment + abandonment.”

3) Triangulation (super testable + easy marks)

Triangulation = don’t trust one method only. You use multiple methods and/or multiple researchers to study the same thing, and if they converge, confidence increases.

Perfect exam phrasing (from your notes):

- “No single method is perfect.”

- “One study is not scientific truth.”
- “Confidence grows when different methods reach the same conclusion.”

Example (use in exam): evaluate a booking/payment flow using (1) usability testing, (2) survey, (3) analytics drop-off; if all point to “payment is confusing,” evidence is strong.

4) Types of research (Descriptive vs Relational vs Experimental)

Your course wants you to remember **the claim type + typical methods**:

Descriptive (“X is happening”)

Goal: describe what’s happening, **no cause-effect**.

Methods: observations, field studies, focus groups, interviews.

Example phrasing: “Users often click back on the payment screen.”

Relational (“X is related to Y”)

Goal: find relationships/correlations; **usually not causation**.

Methods: observations, field studies, surveys.

Example: “Age is related to error rate.”

Experimental (“X is responsible for Y”)

Goal: establish **cause-and-effect** using controlled experiments.

Example: “Adding a summary screen causes fewer payment errors.”

5) Experimental Design (random assignment, types, IV/DV)

Your notes define experiment “properly” using:

- **Random assignment:** if yes → “true experiment”; if no → not a true experiment
 - You also classify: quasi-experiment vs non-experiment depending on whether there are comparisons/measures over time
 - Define **Independent Variable (IV)** = what you manipulate; **levels/values** = versions of IV you compare
 - Define **Dependent Variable (DV)** = what you measure (time, errors, satisfaction)
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6) Other data collection methods (know when to use which)

Your file lists these with “best use cases”:

- **Surveys:** lots of responses fast, geographically dispersed, “big picture.”
 - **Diaries (ethnography-related):** best when things change over time (mood/usage); avoids after-the-fact recall bias.
 - **Case studies:** deep study of one/few real contexts; time-intensive but rich; helps generate hypotheses.
 - **Interviews & focus groups:** direct user opinions/experiences; focus groups add group dynamics.
 - **Ethnography:** observe/participate in natural environment to understand culture/behaviors/needs (what people do vs what they say).
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Part C — Design Rules + Heuristics (memorize lists + give examples)

7) Design rules: Standards vs Guidelines vs Principles

Your lecture’s classification is by:

- **Authority** (must-follow vs suggestion)
- **Generality** (broad vs specific)

So you get:

- **Standards:** high authority, low generality (mandatory + specific; ISO/company design system)
 - **Guidelines:** lower authority, higher generality (recommended practices)
 - **Principles:** abstract + fundamental; interpreted across contexts
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8) Heuristics: what they are

Heuristics are “broad brush” rules used as a **checklist**. They’re general principles that work across many interfaces.

9) Shneiderman’s 8 Golden Rules (MEMORIZE + exam phrasing)

1. **Strive for consistency** (visual + behavioral; reduces cognitive load)
2. **Cater to universal usability** (novices→experts; include shortcuts for experts)

3. **Offer informative feedback** (clear timely system reaction to actions)
 4. **Design dialogs to yield closure** (clear beginning-middle-end; completion confirmation)
 5. **Error prevention + simple error handling** (prevent first; else clear recovery messages)
 6. **Permit easy reversal of actions** (undo/redo → exploration, reduced anxiety)
 7. **Support internal locus of control** (users feel in control, predictable system)
 8. **Reduce short-term memory load** (recognition over recall; chunking; don't make users remember across screens)
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10) Nielsen's 10 Usability Heuristics (MEMORIZE with examples)

1. **Visibility of system status** (feedback; progress bars)
 2. **Match system to real world** (user language, real conventions; shopping cart icon)
 3. **User control and freedom** (undo/redo; "Cancel" exit)
 4. **Consistency and standards** (follow platform conventions)
 5. **Error prevention** (disable submit until required fields filled)
 6. **Recognition rather than recall** (visible options; recently viewed list)
 7. **Flexibility and efficiency of use** (accelerators; shortcuts like Ctrl+S)
 8. **Aesthetic and minimalist design** (remove irrelevant info; uncluttered)
 9. **Help users recover from errors** (plain language; suggest fix)
 10. **Help and documentation** (searchable, task-focused, concrete steps, not too large)
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Part D — Cognition (Week 4) — this is where teachers “nitpick”

Your notes literally give you a “ratta list,” meaning these are high-frequency exam targets.

11) What cognition includes (definition-level)

Cognition = mental processes involved in understanding: **attention, perception, memory, learning, reasoning** (and decision-making).

12) Experiential vs reflective cognition (Norman) + Fast vs slow thinking (Kahneman)

You're expected to contrast these as “two modes”:

- **Experiential**: quick, intuitive, automatic (feels immediate)
- **Reflective**: deliberate, conscious, analytical (thinking it through)
Related: **fast vs slow** thinking; fast = automatic/low effort; slow = effortful/analytical.

13) Attention (focused vs divided) + design implication

Your lecture uses evidence: Tullis results show search is faster when information is **grouped with spacing** (3.2s vs 5.5s).

Also: hands-free driving is still dangerous because the issue is **cognitive distraction**, not just holding the phone.

14) Perception (what it affects in UI)

Design implications list includes things like: icons, spacing, contrast, sound, haptics (your notes flag these explicitly as memorization points).

15) Memory (working vs long-term) + recognition over recall

Your notes say:

- Working memory limits exist (and you must not misuse 7 ± 2)
- Big principle: **recognition is easier than recall** → interfaces should show options rather than make users remember commands.

16) Learning (incidental vs intentional)

Key idea: people learn by doing; allow exploration but provide guidance; differentiate incidental vs intentional learning.

Part E — Cognitive frameworks (mental models, gulfs, distributed/external/embodied)

17) Mental models (definition + famous thermostat example)

Mental model = user's internal understanding of how something works, used to predict outcomes. Users often have incorrect models.

Classic example: thermostat misunderstanding ("higher setting heats faster" — wrong; it's basically on/off).

Design goal: help users form accurate models via clear instructions/tutorials and transparent UI.

18) Gulfs of Execution & Evaluation (Norman) — DEFINITIONS + the questions

- **Gulf of execution:** gap between user goal and actions system allows → user asks: "What can I do? How do I do it?"

- **Gulf of evaluation:** gap between system state and user understanding → user asks: “What happened? Is this what I wanted?”
Good design bridges gulfs by making actions obvious and feedback clear (reduces cognitive effort).

19) Distributed cognition

Cognition isn’t just “in one person’s head.” It’s distributed across:

- people
 - artifacts/tools
 - environment/representations
- Example: airplane cockpit/ATC — safety depends on the whole cognitive system.

20) External cognition (offloading)

External cognition = using external representations (notes, maps, diagrams) to extend thinking.

- **Cognitive offloading:** reminders/to-do lists reduce remembering effort
- **Computational offloading:** representation matters (Roman numerals vs Arabic multiplication example)

21) Embodied interaction

Embodied cognition: thinking is shaped by bodily engagement with the world; touch/VR/tangible interfaces can change understanding.

Part F — Interaction concepts you can be asked to “list + differentiate”

22) Interface metaphors

Metaphor = making an interface resemble a physical entity/activity to use familiar knowledge for unfamiliar functions.

Examples: surfing the web, desktop (files/folders/trash), shopping cart, **card metaphor**.

Benefits:

- Easier learning, helps conceptual model, can aid accessibility/innovation.
Problems:
- Cultural mismatch, constrains thinking, transfers bad designs, doesn’t scale for complexity.

23) Interaction types (list + examples)

Five types (your notes say memorize):

- **Instructing** (commands/options)
- **Conversing** (chat/voice)
- **Manipulating** (direct manipulation: drag/zoom)
- **Exploring** (maps/VR/AR)
- **Responding** (system initiates notifications/alerts)

Direct manipulation (Shneiderman) — 3 properties

1. continuous representation
2. physical actions instead of complex commands
3. rapid reversible actions + immediate feedback

Benefits + disadvantages are also testable.

IMPORTANT distinction your teacher may ask:

- Interaction type = what user is doing conceptually
 - Interface style = the UI form that supports it (menu, CLI, gesture, voice)
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Part G — Navigation + Screen layout + Error reduction (applied answers)

Your Week 6 model answer is basically an exam template.

24) Four “golden rules” of navigation (the “where” questions)

- Show **where you are** (breadcrumbs)
- Show **what you can do** (obvious actions)
- Show **where you are going** (clear next-step labels)
- Show **where you’ve been** (progress indicators / steps)

25) Screen layout principles (the “fix this UI” part)

- **Group related items**
- **Order** content in natural flow; ensure correct tab order
- **Alignment** for scanning (left-align text; align columns)
- **White space** to separate sections, reduce clutter, highlight warnings

26) Designing for human error + reducing cognitive load

Your notes explicitly connect this to:

- Prevent slips with confirmations on dangerous actions (e.g., “Drop course?”)
 - Provide **undo**
 - Reduce reliance on STM by keeping selected items visible (recognition)
 - Use feedback to prevent wrong mental models (mistakes)
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Part H — Visualization, VR/AR/Ubiquitous computing (“Between Two Worlds”)

27) Visualisation types + examples

Your notes cover:

- Scientific/technical data: real space is 3D → use 3D representations (virtual wind tunnel)
- Structured information:
 - **Hierarchies** (file trees/org charts)
 - **Networks/graphs** (web pages/links)
 - Problem: wide hierarchies get huge in 2D; solution: **3D Cone Trees (Xerox)** with rotatable viewing.
- Time & interactivity:
 - map time to space (graphs/timelines), replay events, “Visible Human Project” scrolling slices
 - interactivity: thresholds/filters/zoom; Influence Explorer example

28) “Between two worlds” summary (big concluding idea)

- Ubiquitous computing: computers embedded in real world
 - VR & visualisation: real world represented inside computers
 - AR & ambient displays: physical + digital intermingled
- Maturity: VR/visualisation common; AR/ubicomp rapidly growing.